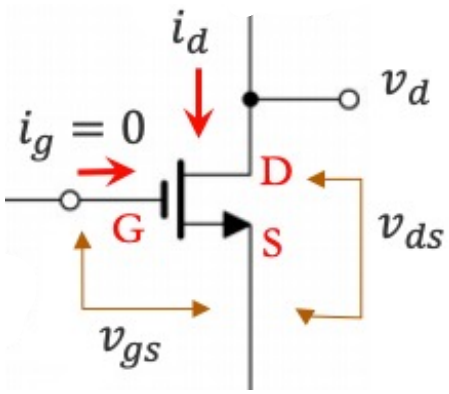
Field Effect Transistors

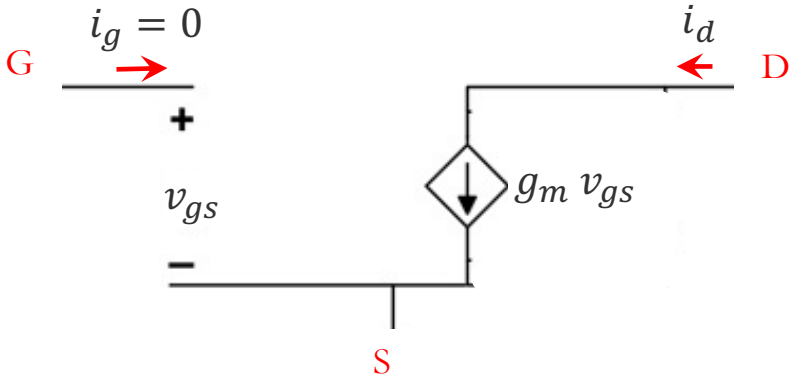
- MOSFET Amplifier Circuit Analysis.

MOSFET Small-signal Circuit Model

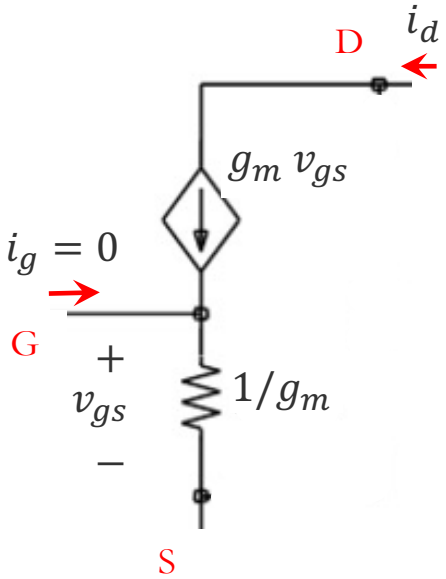


The transistor's small-signal current and voltage relations can be converted to a "circuit model".

$i_g = 0$	➡	Open circuit
$i_d = g_m\, v_i$	➡	Controlled current source
$\lambda \neq 0$	➡	Output resistance $r_o = V_A / I_D$



The NMOS π Model



The NMOS T Model

Small-signal Analysis

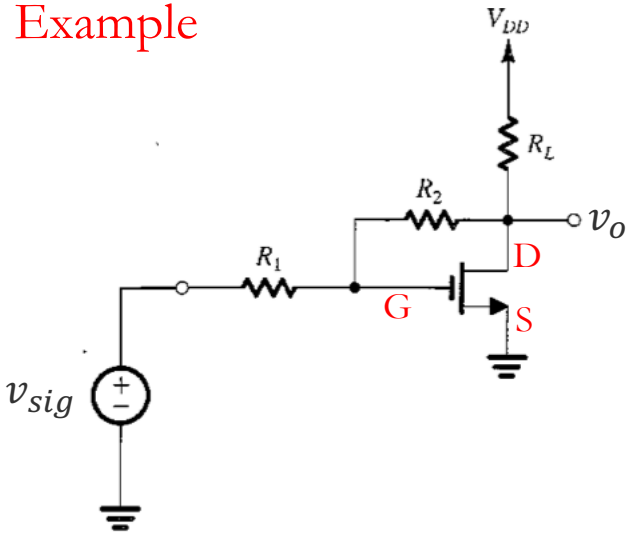
Procedure

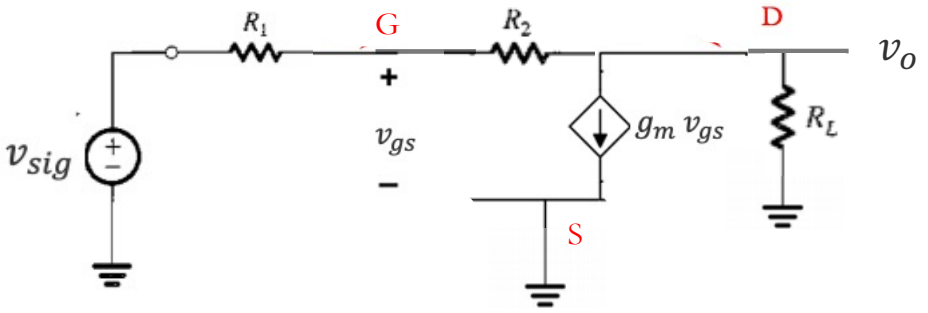
1) Perform DC analysis of the circuit. Find I_D and hence g_m and r_o.

2) Produce the small-signal equivalent circuit using the pi or T model of the transistor.

3) Analyze the circuit using circuit equations (KVL, KCL, etc.) to find important amplifier parameters (gain, input/output resistances, etc.)

Example

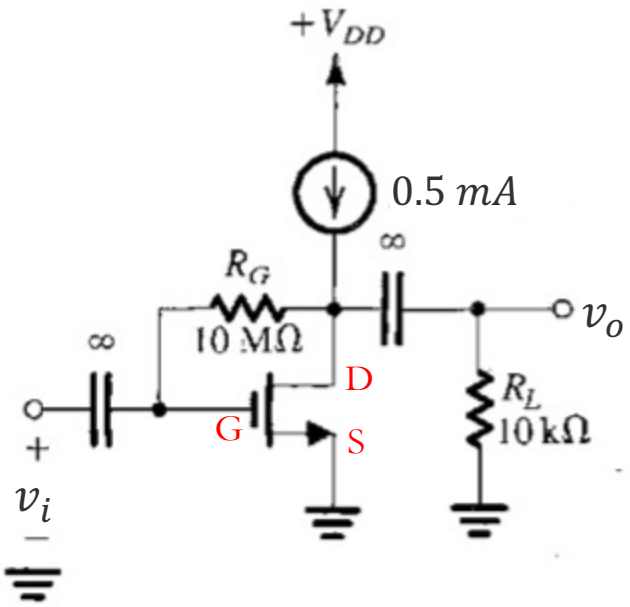




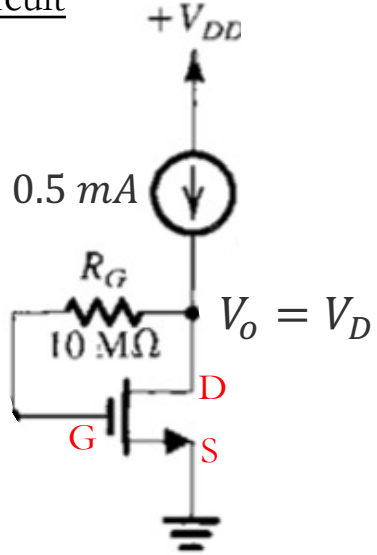
Small-signal equivalent circuit

Example

Find the voltage gain, input and output resistances of the amplifier circuit shown below. Take $V_t = 0.5\ V, V_A = 50\ V$ and $k = 2\ mA/V^2$.



DC Circuit



$I_D = 0.5\ mA$

$g_m = 2\sqrt{k}\sqrt{I_D} = 2\ mA/V$

$r_o = \dfrac{V_A}{I_D} = 100\ k\Omega$

DC Analysis

Transistor is in saturation $(V_D = V_G)$

$I_D = k(V_{GS} - V_t)^2$

$0.5\ m = 2\ m\ (V_D - 0.5)^2$

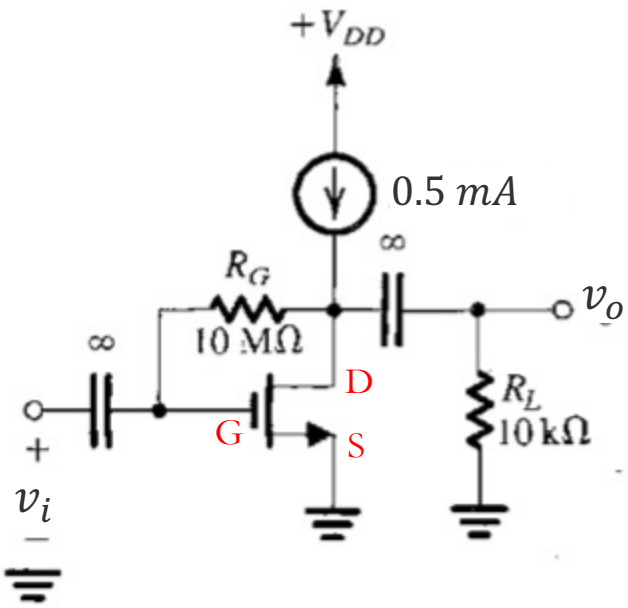
$V_D = +1\ V$

$V_o = +1\ V$

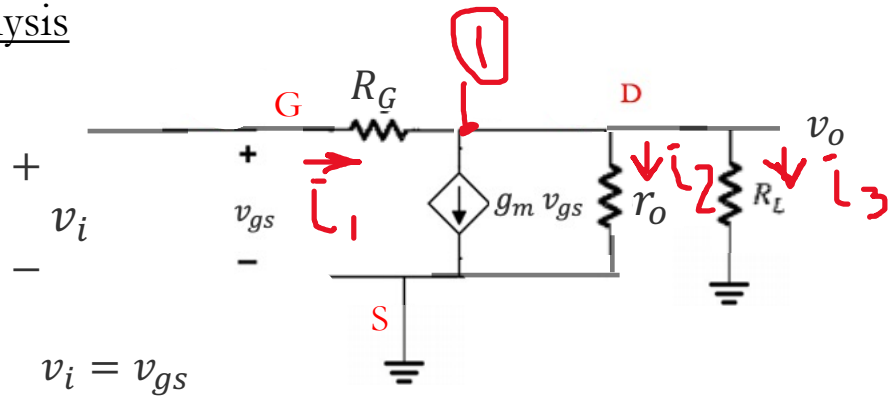
Capacitors protect the DC conditions and clean the ac output.
Coupling and bypass.

Example

Find the voltage gain, input and output resistances of the amplifier circuit shown below. Take $V_t = 0.5\,V, V_A = 50\,V$ and $k = 2\,mA/V^2$.



AC Analysis



$v_i = v_{gs}$

KCL at node 1 $\quad i_1 = g_m v_{gs} + i_2 + i_3$

$$\frac{v_i - v_o}{R_G} = g_m v_i + \frac{v_o}{r_o} + \frac{v_o}{R_L}$$

$$-v_o\left(\frac{1}{R_G} + \frac{1}{r_o} + \frac{1}{R_L}\right) = v_i\left(g_m - \frac{1}{R_G}\right)$$

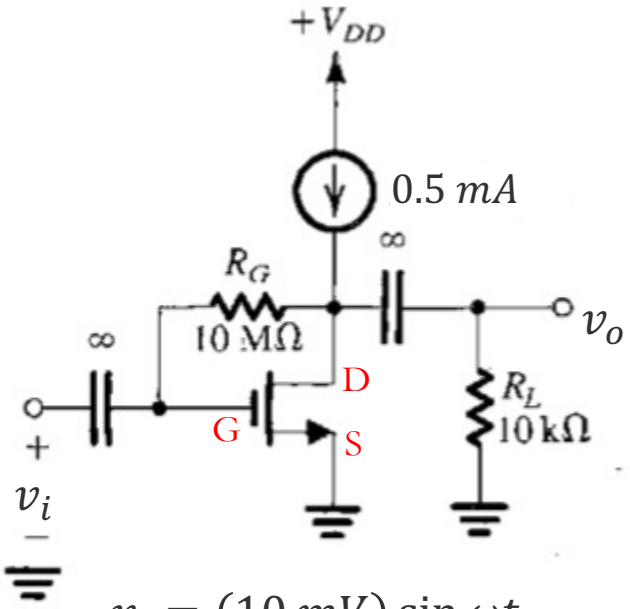
If we ignore r_o,

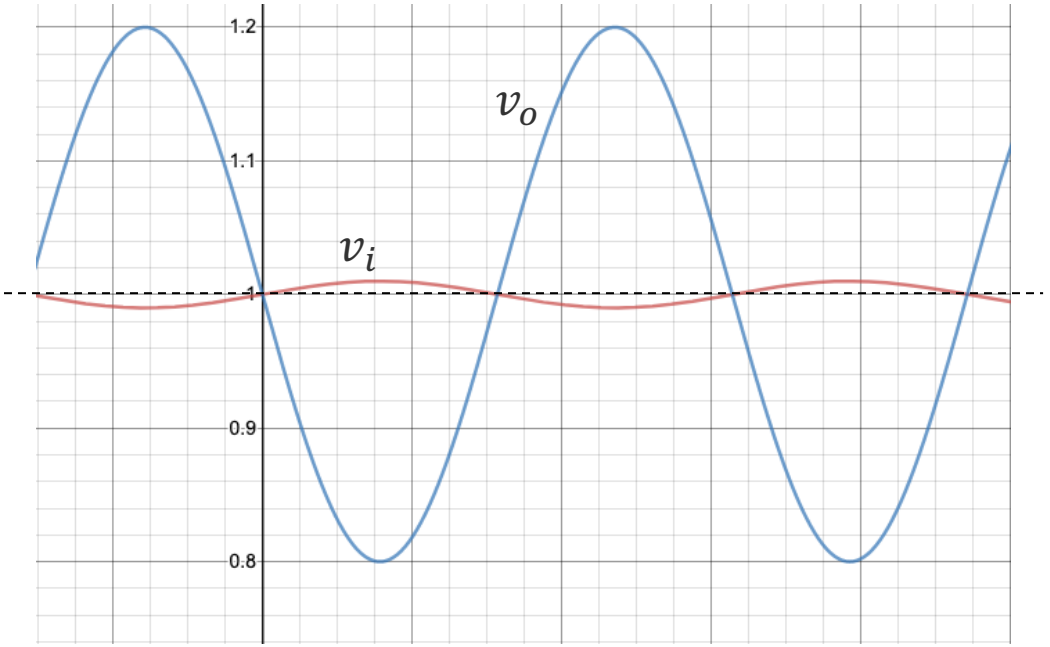
$$A_v = \frac{v_o}{v_i} = \frac{1 - g_m R_G}{1 + R_G/R_L} = -20.0\,V/V$$

$$A_v = \frac{v_o}{v_i} = \frac{1 - g_m R_G}{1 + R_G/(r_o \parallel R_L)} = -18.18\,V/V$$

Example

Find the voltage gain, input and output resistances of the amplifier circuit shown below. Take $V_t = 0.5\ V, V_A = 50\ V$ and $k = 2\ mA/V^2$.





$$v_i = (10\ mV)\sin \omega t$$

$$v_o = (-20)(10\ mV)\sin \omega t$$

Peak voltage at the output (Drain)

$$v_D = 1 \pm 0.2\ (peak)$$

$$v_G = V_G + v_i = 1 + v_i$$

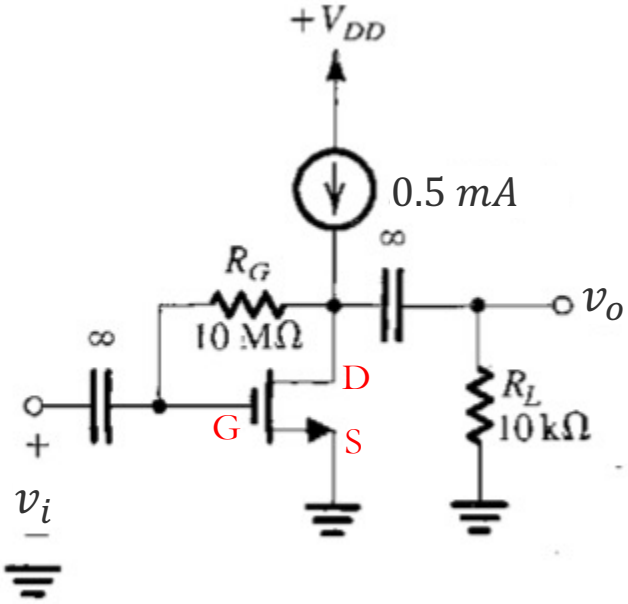
$$v_D = V_D + v_o = 1 + v_o$$

To stay in saturation $\quad v_D \geq v_G - V_t = 0.5\ V$

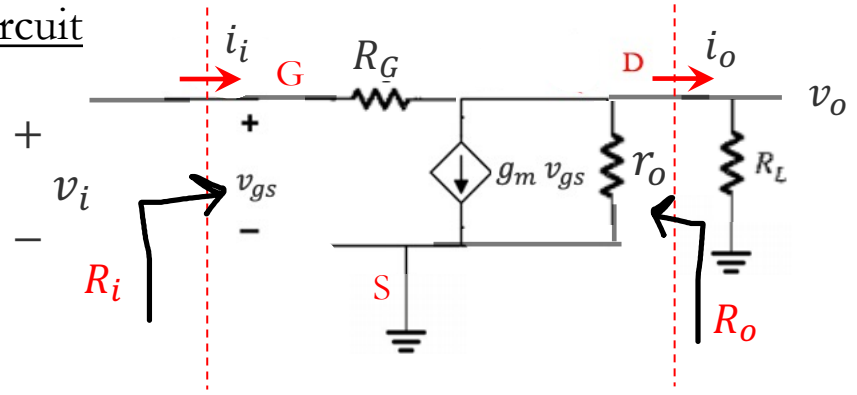
Maximum swing $\quad v_i \leq 25\ mV\ (peak)$

Example

Find the voltage gain, input and output resistances of the amplifier circuit shown below. Take $V_t = 0.5\ V, V_A = 50\ V$ and $k = 2\ mA/V^2$.



AC circuit



To find the equivalent resistance, independent sources need to be properly removed.

Amplifier input resistance $R_i = \dfrac{v_i}{i_i}$

Amplifier output resistance $R_o = \dfrac{v_o}{i_o}$

Find R_i $\quad i_i = \dfrac{v_i - v_o}{R_G} \Rightarrow \dfrac{i_i}{v_i} = \dfrac{1 - v_o/v_i}{R_G}$

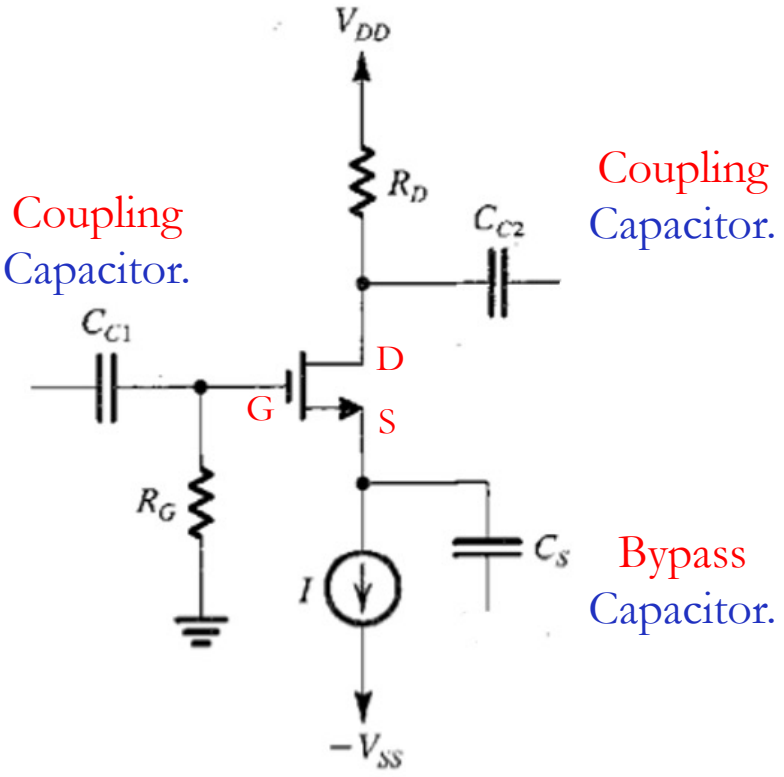
$$R_i = \frac{v_i}{i_i} = \frac{R_G}{1 - A_v} = 0.52\ M\Omega$$

Find R_o $\quad R_o = R_G \parallel r_o \cong 100\ k\Omega$

MOSFET Basic Amplifier Configurations

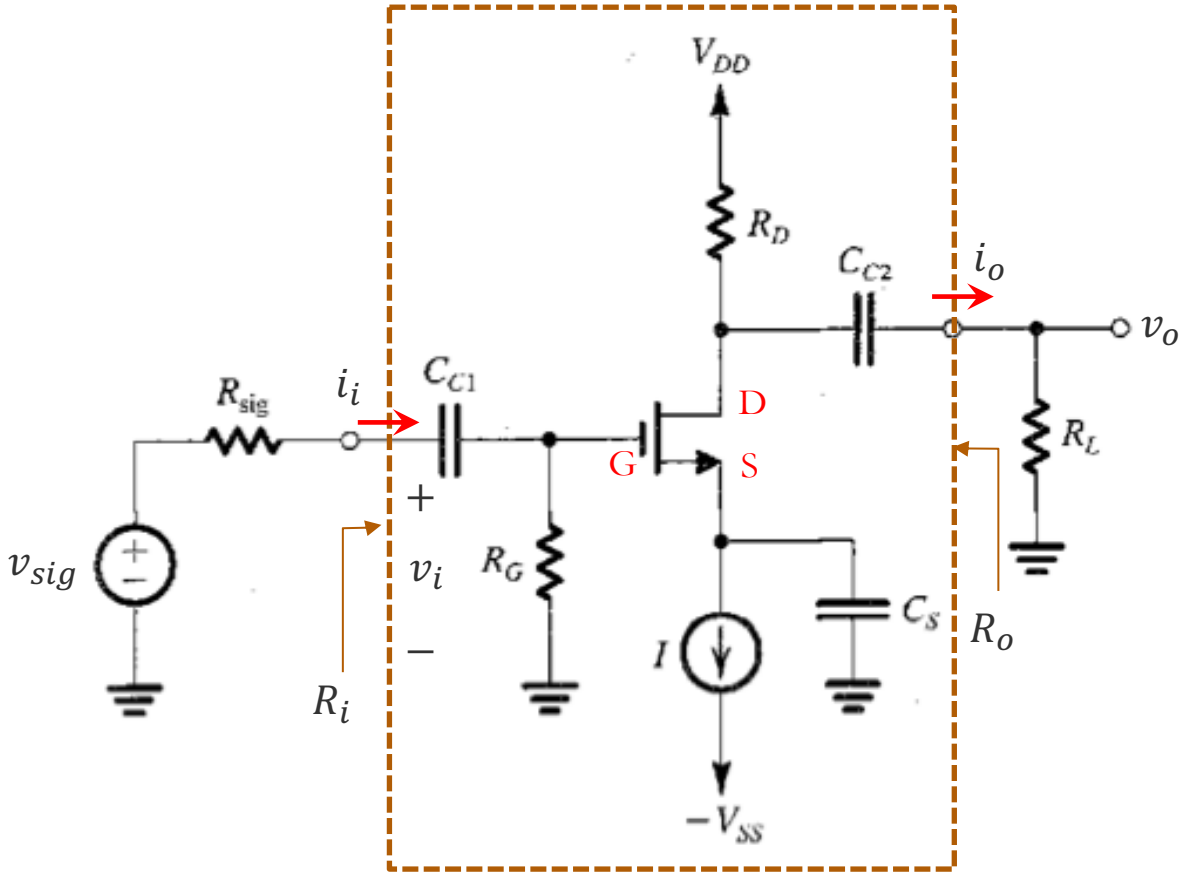
Different combinations of input-output connections are possible; each gives specific amplifier properties.

Input	Output	Ground	Name
Gate	Drain	Source	Common-source (CS)
Gate	Source	Drain	Common-drain (CD)
Source	Drain	Gate	Common-gate (CG)



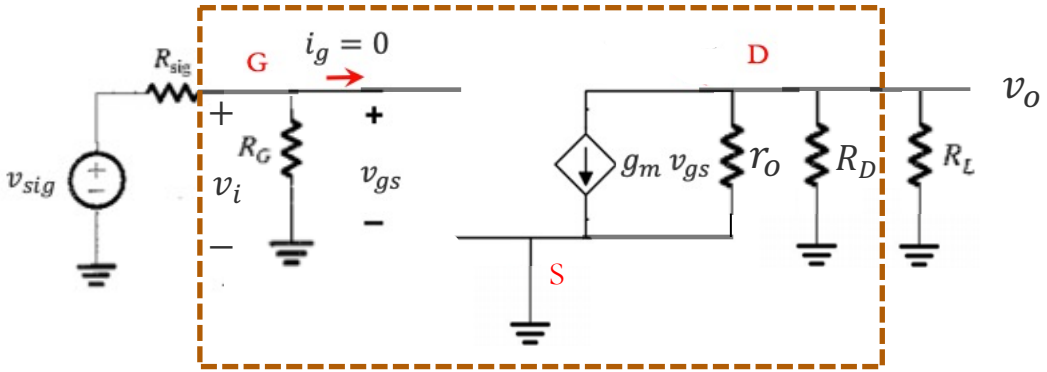
The Common-source (CS) Amplifier

This configuration works as "all-purpose amplifier": high voltage and current gains.



The Common-source (CS) Amplifier

This configuration works as "all-purpose amplifier": high voltage and current gains.



$$v_i = v_{gs} \qquad v_i = v_{sig}\frac{R_G}{R_{sig}+R_G}$$

$$v_o = -g_m v_{gs}\,(r_o \parallel R_D \parallel R_L)$$

$$v_o = -g_m v_i\,(r_o \parallel R_D \parallel R_L)$$

$$A_v = \frac{v_o}{v_i} = -g_m\,(r_o \parallel R_D \parallel R_L)$$

Voltage gain

$$A_v = \frac{v_o}{v_{sig}} = -g_m \frac{R_G}{R_{sig}+R_G}\,(r_o \parallel R_D \parallel R_L)$$

Overall Voltage gain

Open-circuit Voltage gain (no load)

$$A_{vo} = \frac{v_o}{v_i} = -g_m\,(r_o \parallel R_D)$$

Input Resistance

$$R_i = R_G$$

Output Resistance
short-circuit source v_{sig}

$$R_o = r_o \parallel R_D$$

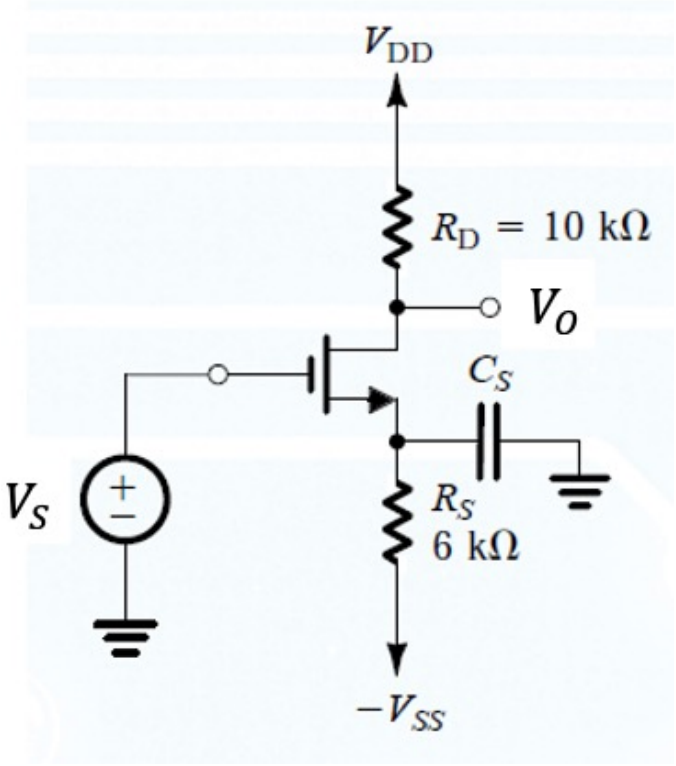
Current gain

$$A_i = \frac{i_o}{i_i} = \frac{v_o}{R_L}\cdot\frac{R_i}{v_i}$$

$$A_i = \frac{i_o}{i_i} = A_v \cdot \frac{R_i}{R_L}$$

Example

Analyze the amplifier given below.



Q & A

Nano Photonics and Plasmonics Research

Send me an email if
you have any
questions.